Section 1: Data Understanding Tasks:

1. Load dataset
2. Display: Shape and Columns
3. Check: Missing values and Data types

```
import pandas as pd # imported pandas and stored it in the variab
df = pd.read_csv('Mall_Customers.csv') # read the csv file
df # get the data
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

Next steps: Generate code with df New interactive sheet

```
df.shape # the number of rows and columns
```

```
(200, 5)
```

```
df.columns # name of columns
```

```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
       'Spending Score (1-100)'],
      dtype='object')
```

```
df.info() # To get the data types
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
```

```
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
1 df.isnull() # to check for the missing values in the data set
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
195	False	False	False	False	False
196	False	False	False	False	False
197	False	False	False	False	False
198	False	False	False	False	False
199	False	False	False	False	False

200 rows × 5 columns

```
1 df.isnull().sum() # To count the number of misisng values from ea
```

	0
CustomerID	0
Gender	0
Age	0
Annual Income (k$)	0
Spending Score (1-100)	0

dtype: int64

```
1 df.isnull().sum().sum() # Count all missing values from the rows
```

```
np.int64(0)
```

Questions:

1. What type of dataset it is?
2. Which features are numerical/categorical?

Ans 1. Type of Dataset: It is a structured tabular dataset that contains customer information.

Ans2. Numerical features are:

1. CustomerID
2. Age
3. Annual Income(k$)
4. Spending Score (1-100)

Categorical features are:

1. Gender

Section 2: Descriptive Analysis Tasks: Compute the mean,median, and standard deviation for:

1. age
2. annual income
3. spending score

```
df['Age'].mean() # to get the mean
```

```
np.float64(38.85)
```

```
df['Age'].median() # to get the median
```

```
36.0
```

```
df['Age'].std() # to get the standard deviation
```

```
13.969007331558883
```

```
df['Annual Income (k$)'].mean() # to get the mean
```

```
np.float64(60.56)
```

```
df['Annual Income (k$)'].median() # to get the median
```

```
61.5
```

```
df['Annual Income (k$)'].std() # to get the standard deviation
```

```
26.264721165271254
```

```
df['Spending Score (1-100)'].mean() # to get the mean
```

```
np.float64(50.2)
```

```
df['Spending Score (1-100)'].median() # to get the median
```

```
50.0
```

```
df['Spending Score (1-100)'].std() # to get the standard deviatio
```

```
25.823521668370162
```

Questions:

1. Which feature shows highest variablity?
2. Are customers similar or diverse?

Ans 1. Highest variablity means we look which feature has the highest standard deviation. Which is 'Annual Income (k$)'.

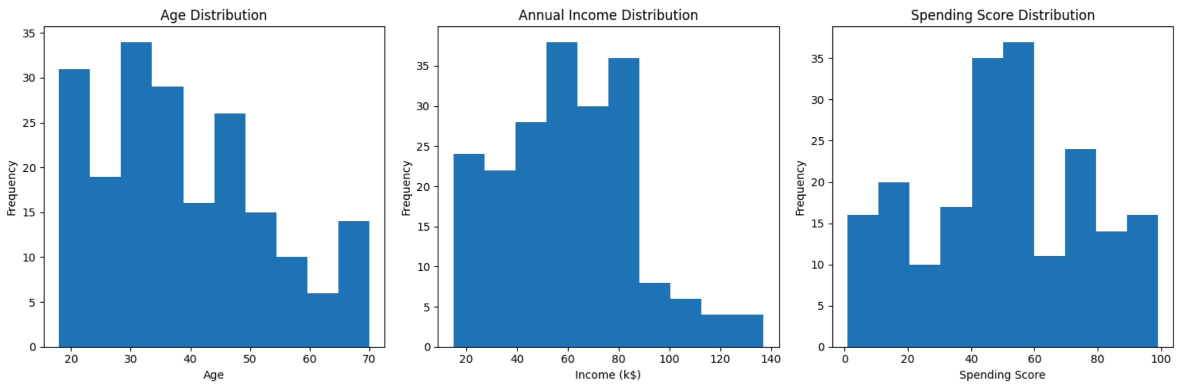
Ans 2. Customers are diverse, as the dataset shows there is significant variability in annual income and spending scores. Which indicates different customer profiles and purchasing behaviors.

Section 3: Visualization Tasks:

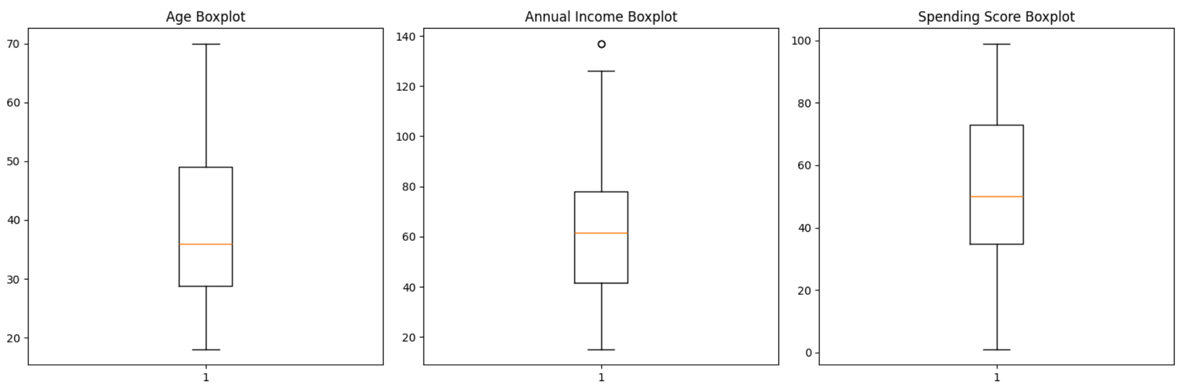
1. Histogram: (Age,Income,Spending)
2. Boxplot: (Detect outliners)

```
import matplotlib.pyplot as plt # Import Matplotlib for data visu
plt.figure(figsize=(15,5)) # Create a figure of size 15x5 inches

plt.subplot(1,3,1) # Create subplot 1 (1 row, 3 columns, position
plt.hist(df['Age'], bins=10) # Plot histogram for Age with 10 bin
plt.title('Age Distribution') # Set title for Age histogram
plt.xlabel('Age') # Label x-axis
plt.ylabel('Frequency') # Label y-axis

plt.subplot(1,3,2) # Create subplot 2 (position 2)
plt.hist(df['Annual Income (k$)'], bins=10) # Plot histogram for
plt.title('Annual Income Distribution') # Set title for Income hi
plt.xlabel('Income (k$)') # Label x-axis
plt.ylabel('Frequency') # Label y-axis

plt.subplot(1,3,3) # Create subplot 3 (position 3)
plt.hist(df['Spending Score (1-100)'], bins=10) # Plot histogram
plt.title('Spending Score Distribution') # Set title for Spending
plt.xlabel('Spending Score') # Label y-axis

```

```
plt.tight_layout() # Adjust spacing between subplots
plt.show() # Display all histograms
```




```
import matplotlib.pyplot as plt # Import Matplotlib for visualiza

plt.figure(figsize=(15,5)) # Create a figure of size 15x5 inches

plt.subplot(1,3,1) # Create subplot 1
plt.boxplot(df['Age']) # Create boxplot for Age
plt.title('Age Boxplot') # Set title

plt.subplot(1,3,2) # Create subplot 2
plt.boxplot(df['Annual Income (k$)']) # Create boxplot for Annual
plt.title('Annual Income Boxplot') # Set title

plt.subplot(1,3,3) # Create subplot 3
plt.boxplot(df['Spending Score (1-100)']) # Create boxplot for Sp
plt.title('Spending Score Boxplot') # Set title

plt.tight_layout() # Adjust spacing between plots
plt.show() # Display all boxplots
```



Questions:

1. Is data skewed?
2. Are there extreme customers?

Ans 1. By looking at the histogram of age distribution it appears to be slightly right - skewed, as there are less customers of higher age. Now, for the annual income distribution it is positively right skewed as there are less customers who earn more. Now for the spending score it appears to be fairly balanced with no strong skewness.

To conclude The dataset shows slight positive skewness, particularly in Annual Income.

Ans 2. Yes. The Annual Income boxplot shows at least one extreme customer with unusually high income compared to the rest of the customers. No significant outliers are observed in Age or Spending Score.

Section 4: Group-based Analysis Tasks:

Compare Male vs Female customers by computing:

1. Average spending score
2. Average income

```
df.groupby('Gender')['Spending Score (1-100)'].mean()    # Calcul
```

Gender	Spending Score (1-100)
Female	51.526786
Male	48.511364

dtype: float64

```
df.groupby('Gender')['Annual Income (k$)'].mean()        # Calcul
```

Gender	Annual Income (k$)
Female	59.250000
Male	62.227273

dtype: float64

Questions:

1. Which group spends more ?
2. Is there a visible difference ?

Ans1 . Female customers spend more on average than male customers, with an average spending score of 51.53 compared to 48.51 for males.

Ans2 . 51.53 - 48.51 = 3.02 There is only a small difference in spending behavior between male and female customers. Female customers have a slightly higher average spending score, but the difference is not very large.

Female Average Income = 59.25 k$MaleAverageIncome = 62.23k$

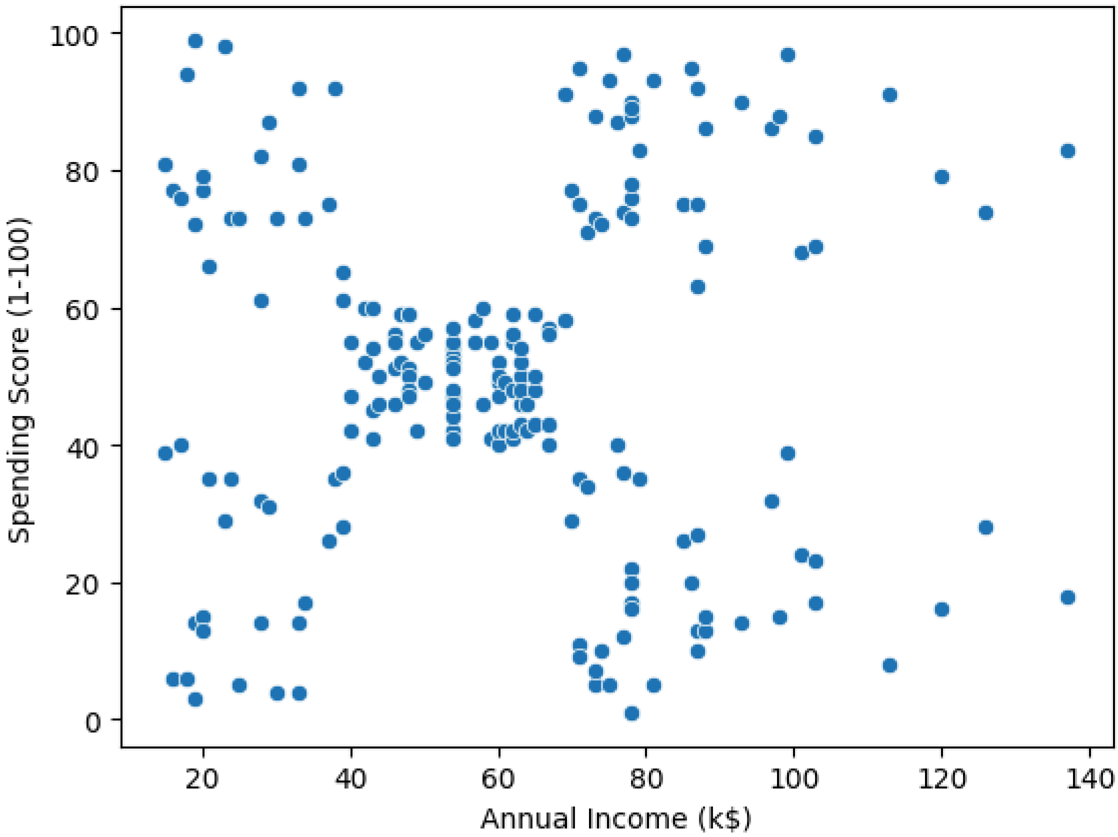
Although males earn slightly more on average, females tend to spend slightly more at the mall.

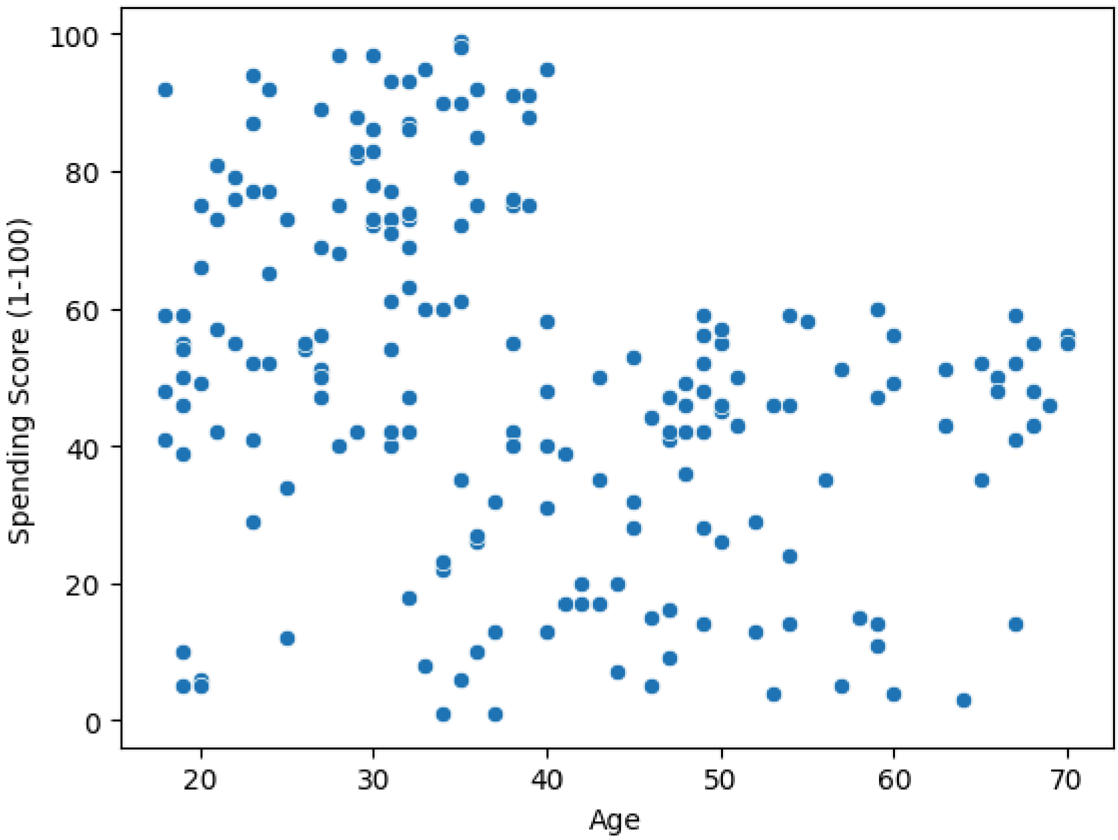
Section 5: Correlation Analysis Tasks:

Identify correlation between:

1. Income vs Spending score
2. Age vs Spending score

```
import seaborn as sns
import matplotlib.pyplot as plt

# Scatter plot: Income vs Spending Score
sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)
plt.show()

# Scatter plot: Age vs Spending Score
sns.scatterplot(x='Age', y='Spending Score (1-100)', data=df)   #
plt.show()
```





Questions:

1. Do richer customers spend more ?
2. Does age affect spending ?

Ans 1. No, higher-income customers do not necessarily spend more. The scatter plot does not show a clear upward trend between Annual Income and Spending Score. Customers with both low and high incomes are distributed across low, medium, and high spending scores. Therefore, income alone is not a strong predictor of spending behavior.

Ans 2. Yes, age appears to influence spending behavior. The scatter plot suggests a negative relationship between Age and Spending Score. Younger customers generally have higher spending scores, while older customers tend to have lower spending scores. This indicates that spending tends to decrease as age increases.

Section 6: Inferntial Statistics(CORE)

Problem Statement: "Do male and female customers spend differently?"

Step 1: Hypothesis H_0: No difference in spending. H_1: There is a difference

Step 2: Perform Test Pythonfrom scipy import statsmale = df[df['Gender'] == 'Male']['Spending Score (1-100)']female = df[df['Gender'] == 'Female']['Spending Score (1-100)']t_stat, p_value = stats.ttest_ind(male, female) Step 3: Decision○ If $p < 0.05 \rightarrow$ Reject H_0

```
from scipy import stats # Import statistical functions from scipy

male = df[df['Gender'] == 'Male']['Spending Score (1-100)'] # Fil
female = df[df['Gender'] == 'Female']['Spending Score (1-100)'] #

t_stat, p_value = stats.ttest_ind(male, female) # Perform indepen

print("T - statistics:", t_stat) # Print t-statistic value
print("P - value:", p_value) # Print p-value result of the test

if p_value < 0.05: # Check if result is statistically significant
    print("Reject Null Hypothesis") # If p < 0.05, there is signi
else:
    print("Accept Null Hypothesis") # If p >= 0.05, no significan
```

```
T - statistics: -0.8190464150660334
P - value: 0.4137446589852174
Accept Null Hypothesis
```

Questions:

Q1. What does p-value tell you?

Q2. Is the difference statistically significant?

Ans1. The p-value tells you how likely it is to get the observed result (or something more extreme) if the null hypothesis is actually true. It helps you understand whether the difference you see in the data is due to random chance or a real effect.

Ans2. No, the difference is not statistically significant because the p-value (0.4137) is greater than 0.05. This means we do not have enough evidence to reject the null hypothesis, so we conclude that there is no significant difference in spending between male and female customers.

Section 7: Confidence Interval

Tasks:

Calculate 95% CI for the spending score

```
import numpy as np # Import NumPy for numerical operations
import scipy.stats as stats # Import statistical functions (not u

# Take the spending score column
spending = df['Spending Score (1-100)'] # Extract spending score 

# Sample statistics
mean = np.mean(spending) # Calculate mean of spending scores
std = np.std(spending, ddof=1) # Calculate sample standard deviat
n = len(spending) # Get total number of observations

# Z value for 95% confidence
z = 1.96 # Critical Z-value for 95% confidence interval

# Standard error
se = std / np.sqrt(n) # Compute standard error of the mean

# Confidence interval
lower = mean - z * se # Lower bound of 95% confidence interval
upper = mean + z * se # Upper bound of 95% confidence interval

print("Mean Spending Score:", mean) # Print mean value
print("95% Confidence Interval:", (lower, upper)) # Print confide
```

```
Mean Spending Score: 50.2
95% Confidence Interval: (np.float64(46.621042491978834), np.float64(5
```

Questions:

Q1: What range do customers fall into?

Q2: How confident are we?

Ans1: The average spending score of customers is 50.2, and the 95% confidence interval is approximately (46.62, 53.78). This means the true average spending score of all customers is expected to lie between 46.62 and 53.78.

Ans2: We are 95% confident that the true population mean spending score falls within this range. This means that if we repeated this sampling process many times, about 95 out of 100 such intervals would contain the true average spending score.

Section 8: Business Insights

Q1. Which customers should the business target?

Q2. Does income affect spending?

Q3. Should marketing differ by gender?

Q4. What strategy would you suggest?

Ans1. The business should target customers based on their spending behavior rather than gender or income. From the analysis, customers with higher spending scores are the most valuable group, so the business should focus more on high-spending customers identified through clustering.

Ans2. No, income does not clearly affect spending. From the scatter plot, there is no strong relationship between annual income and spending score. Customers with both high and low income show mixed spending behavior.

Ans3. No, marketing should not differ by gender. The t-test shows that there is no statistically significant difference in spending between male and female customers, so gender is not an important factor for targeting.

Ans4. The best strategy is to focus on customer segmentation based on behavior. The business should target high-spending customers with premium offers and use

Mall Customers Data Analysis Report

1. Introduction

The objective of this project is to analyze customer data from a mall to understand spending behavior and identify patterns among customers. The dataset includes demographic and spending information.

2. Dataset Overview

The dataset contains 200 records with 5 features: CustomerID, Gender, Age, Annual Income, and Spending Score. There are no missing values.

3. Descriptive Analysis

Average Age: 38.85, Average Income: 60.56k, Average Spending Score: 50.2. Income shows highest variability among features.

4. Data Visualization Insights

Age is slightly right skewed. Income is positively skewed. Spending score is fairly balanced. Few outliers exist in income.

5. Gender Analysis

Females spend slightly more than males (51.53 vs 48.51). Males have slightly higher income (62.23k vs 59.25k).

6. Correlation Analysis

No strong relationship between income and spending. Age shows a negative relationship with spending score.

7. Inferential Statistics

T-test shows p-value = 0.4137, so there is no statistically significant difference in spending between genders.

8. Confidence Interval

Mean spending score = 50.2. 95% CI = (46.62, 53.78). We are 95% confident true mean lies in this range.

9. Business Insights

Focus on customer segmentation based on spending behavior. Gender is not a strong factor. Younger customers tend to spend more.

10. Conclusion

Spending behavior is driven by patterns rather than gender or income. Segmentation-based marketing is recommended.